

Weather Intelligence Knowledge Graph for Pakistan and Neighbouring Regions

PCN Research Internship Assessment 2026

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Executive Summary

This report presents a local weather intelligence knowledge graph built for Task 2 of the PCN Research Internship Assessment 2026. The project addresses the difficulty of analysing weather evidence that is usually separated across daily observations, locations, dates, event summaries and cross-border context. Historical data were collected from the Open-Meteo Historical Weather API for 22 configured locations across Pakistan, India, Afghanistan, Iran and western China/Xinjiang between 1 January 2021 and 31 December 2025. The pipeline cached raw API responses, normalised 40,172 daily records, detected 6,693 weather-event entities using location-month percentile thresholds, and constructed a NetworkX MultiDiGraph containing 12,503 nodes and 45,187 relationships.

The graph represents countries, locations, weather events, dates, time windows and annual climate indicators. A shared graph-query service powers the dashboard, command-line interface and generated analytical exports. It answers six required questions covering rainfall, repeated event exposure, co-occurrence, climate-indicator trends, exposure ranking and cross-border candidate lag patterns. The implementation is reproducible through cached execution, deterministic IDs, validation checks and automated tests. Important limitations are retained throughout: Open-Meteo values are gridded estimates rather than official station records; flood and storm entities are derived candidates; cross-border links do not prove causation or provide forecasts; and the exposure score is not an official vulnerability index.

1. Introduction

Extreme-weather analysis often requires combining several forms of evidence: daily measurements, detected events, geographic context, time windows and possible relationships between events. A flat table can store these values, but it is less convenient for tracing how an event connects to a location, date, indicator or later event. The purpose of this project was therefore to build a reproducible Task 2 pipeline in which weather evidence could be represented and queried as a connected graph.

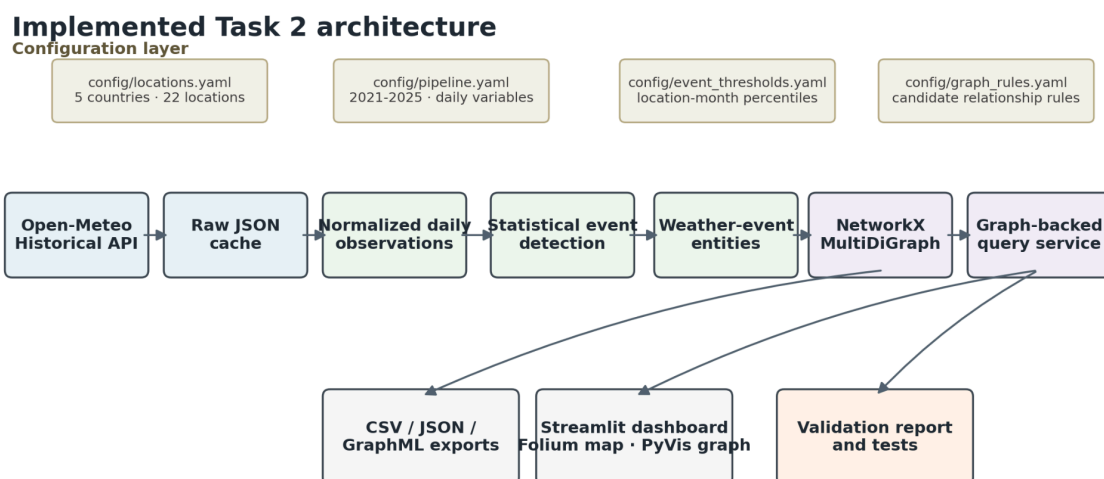
The project only focused on Task 2. Its main goals were to collect weather data for Pakistan and nearby countries, find important weather events, build a knowledge graph, keep clear source details, and answer the six questions given in the task.

The project was made to run in a normal Python 3.11+ setup without Neo4j, Docker, paid tools, or API keys. It was tested on macOS, but it can also run on Windows with the correct Python commands.

The main contributions are an end-to-end cached data pipeline, conservative event-detection rules, deterministic entity and relationship identifiers, a validated NetworkX MultiDiGraph, six shared graph-backed analytical operations, a local Streamlit dashboard, saved Folium and PyVis visualisations, and automated validation.

2. System Architecture and Data Sources

The implemented architecture is shown in Figure 1. Configuration files define the location registry, date range, event thresholds and graph rules. The collector requests historical daily observations from Open-Meteo and stores successful raw responses before any transformation. Normalisation creates a consistent daily schema, after which statistical event detection produces structured event records. These records are converted into a directed multigraph, queried through a shared graph service, exported in several machine-readable formats and presented through local visualisations.



Verified outputs: 40,172 daily rows · 6,693 events · 12,503 nodes · 45,187 relationships

Caveat: all analysis is historical and graph-backed; associations are not proof of causation or forecasts.

Figure 1. Implemented Task 2 architecture from Open-Meteo collection to graph-backed analysis and validation.

Open-Meteo was selected because its historical endpoint supports latitude-longitude queries without registration or an API key [1]. The configured scope contains eight Pakistani locations, four Indian locations, four Afghan locations, three Iranian locations and three locations in western China/Xinjiang. The study period covers 2021-2025. The collected variables include precipitation, rainfall, temperature maxima/minima/means, wind speed, wind gusts, precipitation hours and weather codes.

Raw caching was treated as part of the research provenance rather than only as a performance optimisation. Compatible successful cache files can be reused, cache-only execution prevents unexpected internet calls, and failed responses do not overwrite successful data. The collector also handles HTTP 429 rate limits with bounded retries and waiting logic. The normalised table preserves configured coordinates separately from Open-Meteo grid coordinates because a requested city point may map to a nearby model grid cell.

The resulting dataset contains 40,172 daily rows, equal to the expected coverage, with no duplicate location-date records. These values are gridded historical estimates. They must not be described as official weather-station observations or official meteorological records.

2.1 Project Scope and Verified Totals

Item	Verified value
Period	1 Jan 2021 - 31 Dec 2025
Countries	Pakistan, India, Afghanistan, Iran, China/Xinjiang
Configured locations	22
Normalised daily records	40,172
Detected weather events	6,693
Event thresholds	1,848
Graph size	12,503 nodes; 45,187 relationships
Representative graph	214 nodes; 413 relationships
Validation	100 passed; 0 failed; 2 warnings
Tests	95 passed in latest recorded run

3. Weather-Event Extraction and Entity Resolution

Event extraction uses location-month percentile thresholds instead of one universal threshold. This design accounts for seasonal and geographic differences: a value that is extreme in one location and month may be normal elsewhere. A minimum sample count is required before thresholds are calculated, and missing observations are excluded rather than replaced with zero. Consecutive qualifying dates are grouped into one event.

The detector creates rainfall, temperature, wind and heatwave entities, together with meteorological drought indicators, storm candidates and inferred flood-risk candidates. Storm candidates require compatible rainfall and wind evidence at the same location within a short configured window. Drought indicators use low rolling precipitation, while flood-risk candidates

use unusually high rolling precipitation. Rolling-window start, end and precipitation values are retained so the reported event can be traced to its supporting observations.

3.1 Event Detection Summary

Entity	Detection principle	Interpretation caveat
Rainfall Event	Daily precipitation above location-month 95th percentile and at least 10 mm	Derived event; not an impact report
Temperature Event	High or low temperature beyond local monthly percentile	Derived from gridded temperature
Heatwave	At least 3 days above relative and absolute heat rules	Historical candidate; no attribution
Wind Event	Maximum wind above location-month 95th percentile	Gridded wind estimate
Storm Candidate	Compatible rainfall and wind events within 1 day	Algorithmically derived; not confirmed
Drought Indicator	30-day precipitation below local 10th percentile for at least 10 days	Meteorological indicator only
Flood-risk Candidate	3-day precipitation above local 99th percentile	Rainfall-based inference; not confirmed

Raw severity values are preserved in their original units, but millimetres, degrees Celsius and kilometres per hour cannot be compared directly. A bounded severity percentile is therefore calculated within each event type or subtype. This is a relative analytical ranking, not an official disaster-severity scale.

Entity resolution in this project is deterministic rather than fuzzy. Locations use a controlled registry and stable identifiers such as `pk_karachi`. Event IDs are hashes of the event type, location, subtype and time window; storm IDs additionally include their supporting rainfall and wind events. Country, date, time-window, climate-indicator and relationship IDs are created from canonical attributes. Current integrity checks report no duplicate node IDs, duplicate relationship IDs, dangling endpoints or self-loops.

4. Knowledge Graph Design

The graph is implemented as a NetworkX MultiDiGraph. A directed multigraph is appropriate because relationships have direction and the same pair of entities may be connected in more than one way [2]. For example, an event may both OCCUR_IN and AFFECT a location, while separate event-to-event relationships can express ordering, derivation or candidate cross-border association.

Required node types include Country, Location, Rainfall Event, Temperature Event, Wind Event, Heatwave, Drought, Flood, Storm, Date, Time Window and Climate Indicator. Helper temporal nodes make event windows explicit, while annual Climate Indicator nodes support trend analysis. The required relationships are OCCURRED_IN, AFFECTED, ASSOCIATED_WITH, CAUSED,

PRECEDED, FOLLOWED and UPSTREAM_OF. The implementation also includes LOCATED_IN, STARTED_ON, ENDED_ON and WITHIN_TIME_WINDOW.

Knowledge-graph schema

Main Task 2 entity groups and relationship semantics

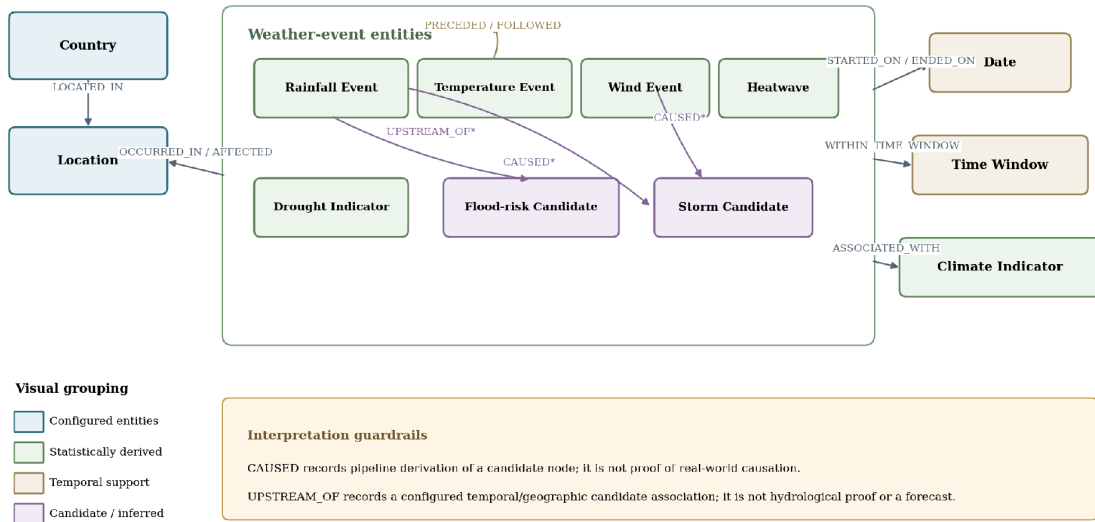


Figure 2. Main entity groups and relationship semantics in the weather intelligence knowledge graph.

Two relationship names require particular caution. CAUSED is used only for pipeline derivation, such as rainfall and wind evidence supporting a storm candidate. It does not claim proven meteorological causation. UPSTREAM_OF links a non-Pakistan source event to a later Pakistani event when the configured corridor, event mapping and lag rules match. It represents a candidate temporal/geographic association, not proof of hydrological flow, a causal conclusion or a forecast.

4.1 Relationship Interpretation

Relationship	Direction	Interpretation limit
OCCURRED_IN / AFFECTED	Event -> Location	Locates the analysis unit; does not quantify damage or population impact
ASSOCIATED_WITH	Event -> Climate Indicator	Analytical association, not causal proof
CAUSED	Rainfall/Wind -> Storm candidate	Pipeline derivation only
PRECEDED / FOLLOWED	Event -> Event	Bounded temporal ordering; not causation
UPSTREAM_OF	Foreign event -> later Pakistan event	Candidate corridor/lag association; not a forecast
STARTED_ON / ENDED_ON	Event -> Date	Explicit temporal provenance

Relationship	Direction	Interpretation limit
WITHIN_TIME_WINDOW	Event -> Time Window	Connects event to its analysed window

Graph growth is controlled through bounded rules. Each event has one main location link; temporal ordering uses nearest-event logic instead of all-to-all links; and cross-border links are restricted to configured corridors, compatible event types and lags of 1-10 days. The generated graph contains 12,503 nodes and 45,187 relationships across five countries. It exceeds the assessment minimums while remaining reproducible from the generated event data.

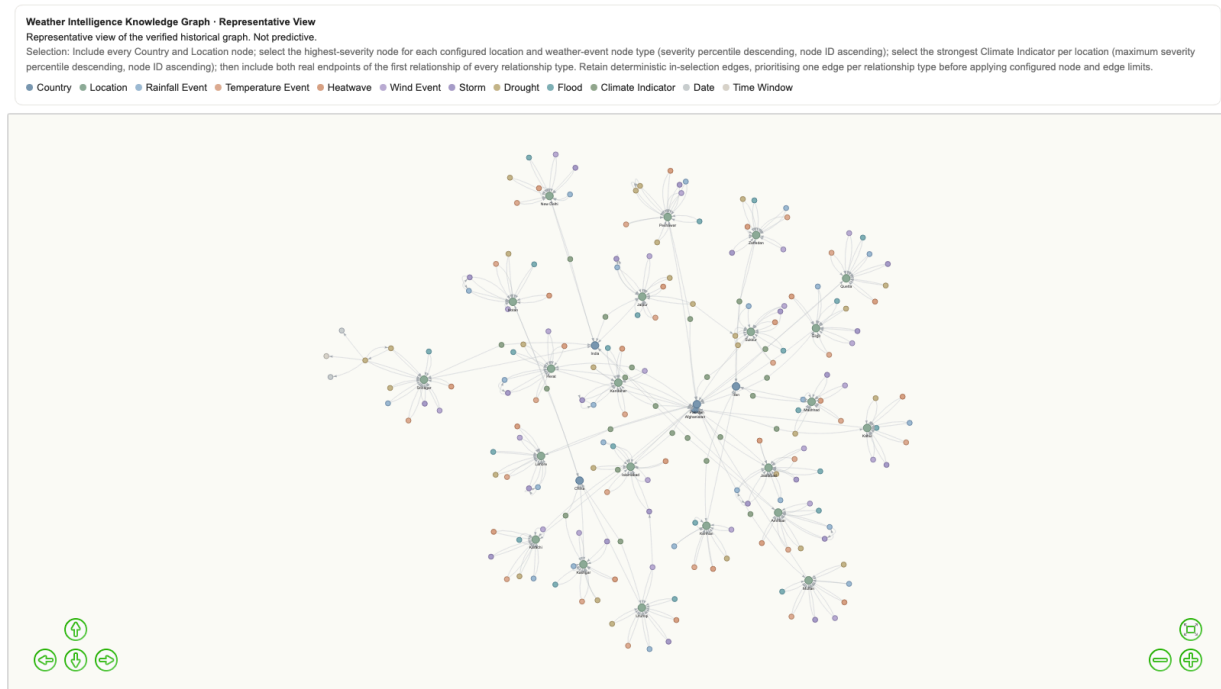


Figure 3. Deterministic representative PyVis view. The analytical backend remains the full GraphML graph.

Graph visualisation note

The representative graph contains 214 nodes and 413 relationships. It is intentionally smaller than the full 12,503-node graph so that locations and selected event evidence remain readable.

5. Graph-Backed Analytical Queries and Findings

The six required analytical operations are implemented in a shared query service that loads the full GraphML file, normalises graph attributes and builds reusable indexes. The dashboard, command-line interface and regenerated analytical CSVs use the same functions. The query categories are predefined by the assessment, but their answers are calculated from the current graph and user-selected filters. The representative PyVis view is not the analytical backend.

During the demonstration, at least two questions can be run live against the constructed graph. Query provenance records the graph path, checksum, parameters, execution duration, inspected nodes or edges and interpretation caveats. This is stronger evidence of live execution than showing a static network screenshot.

5.1 Selected Findings

Query	Current result	Required caveat
Highest rainfall	Karachi Open-Meteo grid point: 162.3 mm on 24 July 2022	Highest value in the analysed Open-Meteo dataset; not an official station record.
Multiple event types	Islamabad: 7 types, 358 events	Configured locations only
Co-occurrence	Temperature + Wind: 978 pairs	Association, not causation
Trend	Gilgit Wind slope: 6.8	Five-year dataset pattern only
Exposure	Islamabad: 0.871901	Weather-event exposure, not vulnerability
Cross-border lag	Kabul -> Peshawar Temp->Temp: 55; median 3 days	Candidate association, not forecast

The main findings are summarised in Table 4 and Figure 4. In the Open-Meteo Historical Weather API data, the selected Karachi grid point returned 162.3 mm of daily precipitation for 24 July 2022 using the Asia/Karachi timezone. The hourly values also sum to 162.3 mm. However, this is a gridded reanalysis/model estimate and should not be treated as an official Karachi weather-station record.

Islamabad contains all seven detected event types and the largest total event count, 358, followed by Gilgit and Lahore. Temperature and Wind is the most frequent co-occurring pair, with 978 pairs across 22 locations and a median gap of one day. Co-occurrence expresses repeated temporal and location association, not causation.

The strongest positive annual event-count slope is Gilgit Wind at 6.8, increasing from one detected event in 2021 to 22 in 2025. These five-year patterns are useful for identifying changes inside the dataset, but they cannot establish long-term climate change or attribution.

The exposure ranking combines frequency, event-type diversity, mean severity percentile and recurrence with equal weights. Islamabad has the highest score, 0.871901. This is a weather-event exposure score only; it does not include population, poverty, infrastructure, health, damage or preparedness and is therefore not an official vulnerability index.

The most repeated cross-border candidate pattern is Kabul to Peshawar for Temperature-to-Temperature events, with 55 UPSTREAM_OF relationships and a median lag of three days. Similar patterns occur from Jalalabad to Peshawar and from Srinagar or Amritsar to Pakistani locations. These outputs are historical candidate associations generated by corridor and lag rules. They are not operational warnings or forecasts.

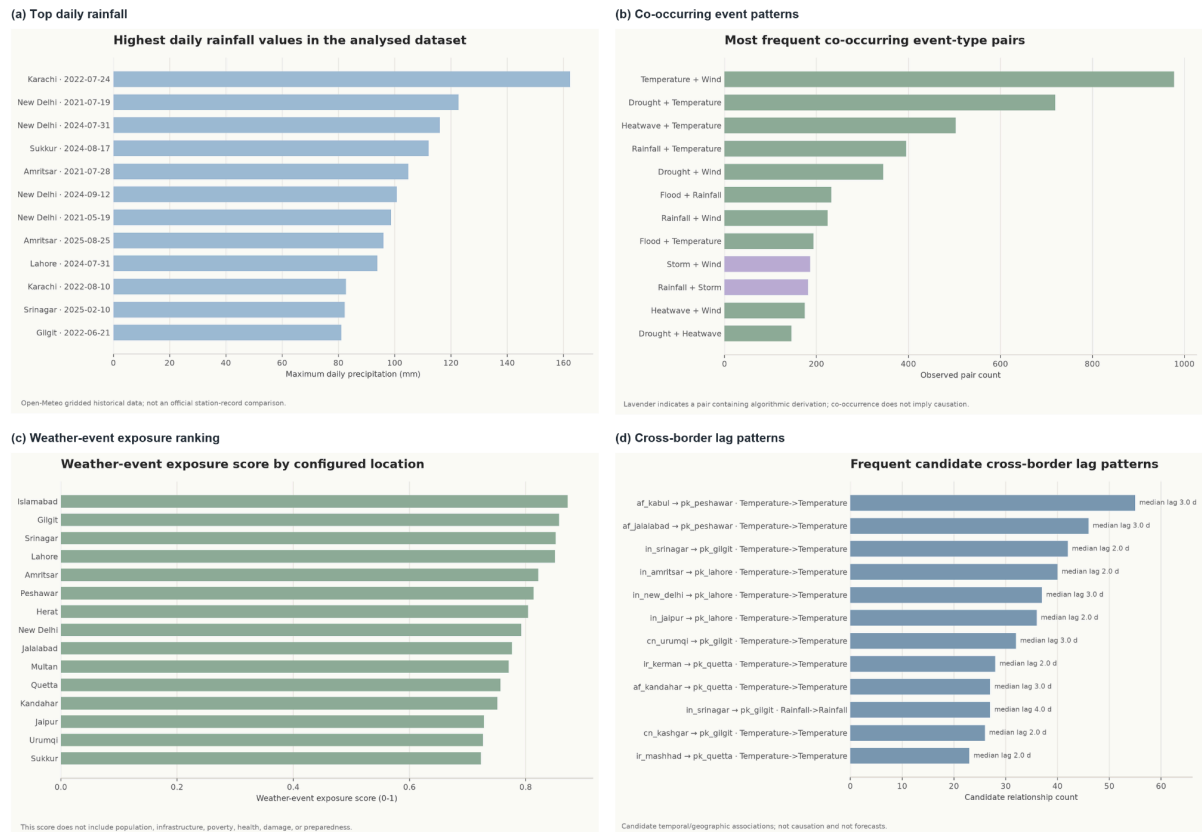


Figure 4. Selected outputs from the rainfall, co-occurrence, exposure and cross-border graph-backed queries

(Rainfall values shown in this figure are Open-Meteo gridded estimates.)

6. Conceptual Task 3 Pipeline - Not Implemented

The submitted repository implements Task 2 only. This short conceptual section is included because the overall report checklist also requests Task 3 pipeline decisions. No Kafka broker, producer, consumer, streaming topic or dead-letter queue was implemented.

Conceptual Task 3 Pipeline - Not Implemented

Included only because the report checklist requests Task 3 pipeline decisions.

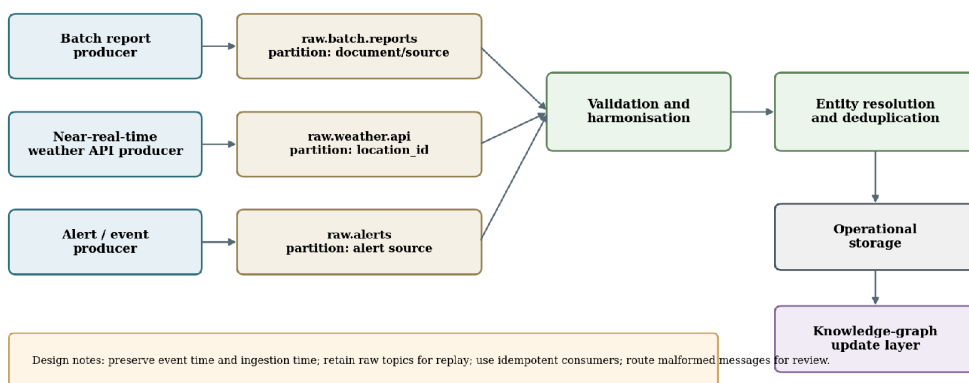


Figure 5. Conceptual Task 3 ingestion pipeline. This design was not implemented in the Task 2 repository.

The aim of this proposed pipeline is to collect new weather information from different sources and safely add it to the Task 2 knowledge graph. The data could come from stored reports, live weather APIs, or warning alerts. Each type of data would be kept in a separate message channel so that it remains organised.

Before the data is added to the graph, the system would check it, clean it, and convert it into one standard format. This format would include details such as the event type, location, source, event time, ingestion time, provenance, and confidence level. Fixed IDs would be used so that the same event is not added more than once.

Incorrect or incomplete messages would be moved to a separate area for review, while the original messages would be saved so they could be processed again if needed.

Batch processing would handle many saved reports together. It would be slower, but easier to check and repeat. Real-time processing would handle weather updates as they arrive. It would provide faster information, but it would be harder to manage because messages may arrive late, arrive in the wrong order, or be received more than once.

After the data is checked and organised, the final event records could be added to the Task 2 graph as new weather-event nodes and relationships. In this way, Task 3 would provide the data-processing pipeline, while Task 2 would store and connect the cleaned weather information. This design is only a future proposal and was not implemented in the submitted project.

7. Challenges, Limitations and Lessons

Several implementation challenges changed the final design. Open-Meteo rate limiting showed why raw caching, respectful request delays and bounded retries are necessary. Early normalisation work also showed that apparently small provenance fields, such as weather code and API grid coordinates, matter later when results are reviewed.

Event labels required repeated interpretation checks. Early storm candidates could be supported by weak rainfall if wind alone was extreme, so the rule was hardened to require compatible rainfall and wind evidence. Raw severities could not be ranked together because they use different physical units, which led to the addition of event-type severity percentiles. Rolling precipitation events required explicit source windows so that drought and flood-risk candidates remained traceable.

I initially assumed that relationship names such as `UPSTREAM_OF` would be self-explanatory. After reviewing the outputs, I realised the name could easily be read as causal or hydrological. The relationship properties, dashboard wording and report therefore describe it as a candidate temporal/geographic association. The Karachi result also showed the importance of describing the data source clearly. The value comes from an Open-Meteo grid point, so I reported it as a result from the analysed dataset rather than as an official weather-station record.

The project remains limited by its five-year period, 22 configured locations, gridded data source and statistically derived event definitions. It does not validate events against official disaster reports, use river-flow or soil-moisture observations, model social vulnerability, forecast future events or perform climate attribution. The PyVis overview is a representative subset, not a display of every node and relationship.

8. Use of LLM Tools and Student Contribution

LLM tools were used during the project for development support, technical explanations, testing assistance, documentation and report preparation. OpenAI Codex assisted with generating and editing parts of the Python implementation, tests, command-line tools, dashboard components, visualisations and validation checks. ChatGPT helped break the assessment into stages, explain concepts, review methodological choices, identify possible overclaims, prepare verification steps and organise the report and demonstration.

My contribution included selecting Task 2, defining the geographic and temporal scope, approving the data source, reviewing event-detection rules, making decisions about graph semantics and interpretation, running commands and tests, inspecting generated files, checking the Karachi Open-Meteo value and how it should be interpreted, reviewing dashboard and visualisation output. Numerical findings were obtained from the project's generated outputs, and the final interpretation and submission decisions remained my responsibility along with the written report and documentation.

9. Conclusion

The project demonstrates an end-to-end method for transforming historical weather observations into a queryable knowledge graph. It combines cached data collection, seasonal percentile-based event extraction, deterministic graph construction, reproducible analytics and local visual review. The graph structure makes provenance and relationships explicit while allowing the same analytical logic to support dashboard, CLI and exported results.

The findings show how a graph can organise rainfall extremes, repeated event types, co-occurrence, short-term indicator trends, exposure components and cross-border lag candidates. Their value depends on careful interpretation: the system provides historical analytical evidence, not official records, disaster confirmation, causal proof or forecasts. Future work could extend the time period and location coverage, integrate official station and disaster reports, add hydrological and socioeconomic data, improve spatial relationships and evaluate an operational streaming design.

References

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